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Global Facilities - Design & Construction Standard - Mechanical - Electrical / Utilities Area / Warehouse GHVAC System

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1.0 Purpose

- 1.1 This document serves as a System Standard document that provides specifications that shall be followed during the Design, Engineering, and Construction of new Electrical / Utilities Area / Warehouse GHVAC Systems as part of new Greenfield FAB Construction projects.
- 1.2 The standard is applicable for GHVAC Systems provided in Electrical Rooms / Utilities Areas / Warehouse that accommodate electrical, utilities equipment and finished goods.
- 1.3 The document is intended to supplement the actual Construction Drawings and Construction Specifications for Construction projects.

2.0 Scope

- 2.1 This document covers the new GHVAC systems serving Electrical Rooms / Utilities Rooms / Warehouse as part of new Expansion Construction projects. This document may or may not apply to existing building space renovation and retrofits of existing Sites, depending on the limitations of the existing Design Concept and System Configuration of those Sites and buildings.
- 2.2 This document applies to all Micron Facilities employees, contractors, and vendors at all Micron sites.
- 2.3 This document covers general utilities rooms only, and does not cover any Hazardous Production Material (HPM) rooms, such as for the storage or dispensing of chemicals or gases.

3.0 Definitions and Acronyms

- **AC** – Air conditioned
- **AHJ** – Authority having Jurisdiction
- **AHU** – Air Handling Unit
- **CDA** – Clean Dry Air
- **CHW** – Chilled Water
- **CUB / CUP** – Central Utilities Building / Central Utilities Plant
- **DOL** – Direct On-Line
- **FCU** – Fan Coil Unit
- **FG** – Finished Goods
- **FMCS/SCADA** - Facility Monitoring and Control System / Supervisory Control and Data Acquisition
- **GHVAC** – General Heating, Ventilation, Air Conditioning
- **HV / MV / LV** – High Voltage / Medium Voltage / Low Voltage
- **LCP** – Local Control Panel
- **LOTO/COHE** – Lockout Tagout / Control of Hazardous Energy
- **MAU** – Make up Air Unit
- **MUA** – Make up Air
- **MV** – Mechanical Ventilation
- **NV** – Natural Ventilation
- **N+1+F** - System sizing concept for equipment quantity consisting of N duty units, +1 redundant standby unit, + future unit(s)
- **OA** – Outside Air
- **PLC** - Programmable Logic Controller
- **POC/LPOC** - Point of Connection / Lateral Point of Connection
- **POU** – Point of Use
- **RA** – Return Air
- **SA** – Supply Air
- **UPS** - Uninterruptible Power Supply
- **UPW** – Ultra Pure Water
- **VAV** – Variable Air Volume Device
- **VSD/VFD** - Variable Speed Drive / Variable Frequency Drive
- **WWTP** – Waste Water Treatment Plant

4.0 Roles and Responsibilities

	Responsibilities
Global Facilities Engineering Team - Global (Discipline Engineer)	• Develop and maintain System Standard document • Ensure Best Known Methods (BKM), lessons learned, and emerging technologies are documented in System Standard • Perform periodic document review and update System Standard
Global Facilities Engineering Team - Regional (Discipline Engineer)	• Review and provide technical feedback on System Standard • Support the Site teams by providing knowledge and technical expertise
Site Project Construction Teams	• Review and provide technical feedback on System Standard • Incorporate System Standard into Project Engineering, Design, and Construction • Document project deviations, exceptions, and exclusions from System Standard • Share new lessons learned to Global Facilities Engineering Team, to determine if it is Best Known Method (BKM) that can be captured in future revision of System Standard
Site Facilities Teams	• Use System Standard as technical reference

5.0 Overview of Systems

The following spaces and systems are used to support the requirements of the building's ventilation air and heating / cooling loads:

- 5.1 **Central Utilities Building / Central Utilities Plant (CUB / CUP)** accommodates the equipment that provides utilities to the production floors and the entire area of the Fab, to support the wafer production and the operation of the buildings.
- 5.2 **Electrical Rooms** accommodate the electrical equipment or devices (i.e., switchgears, transformers, electrical panels, etc), which regulate, supply and isolate electrical power to the building.
- 5.3 **Utilities Rooms** accommodate the equipment or installation (i.e. chillers, compressors, pumps, tanks, etc) that generate and supply utilities (i.e. CHW, UPW, CDA, etc), to support the wafer production and the operation of the Fab buildings.
- 5.4 **Warehouse** accommodate the semi-finished goods and finished goods that have come off the production lines and are ready to be further process, sold or distributed.
- 5.5 **Makeup Air (MUA)** is 100% outside air (OA) ventilation that is supplied to provide outdoor air ventilation for occupants to meet Indoor Air Quality requirements, and it also provides control of indoor air temperature, humidity, and room pressurization level. It compensates for the exhausted air via various exhaust systems or natural exfiltration. The air is conditioned, filtered, and supplied via the Makeup Air Units (MAU).
- 5.6 **Return Air (RA)** is the air that is returned back to the HVAC system from the AHU / FCU units after it has been supplied to each room. After returning to the HVAC system, this Return Air is typically mixed with Outside Makeup Air to provide the Supply Air.
- 5.7 **Supply Air (SA)** is the air that is supplied through the HVAC system from the AHU / FCU units to each room, to regulate the room air temperature, humidity, and indoor air quality. This Supply Air is typically a mixture of Return Air and Outside Makeup Air.
- 5.8 **General HVAC (GHVAC)** System is the air handling system used to control the temperature, humidity, and cleanliness of the air in an enclosed space, in order to provide thermal comfort and acceptable indoor air quality. "HVAC" stands for Heating, Ventilating, Air Conditioning.
- 5.9 **General HVAC (GHVAC) Exhaust** is used to remove non-corrosive, non-toxic, non-hazardous, or smoke-logged indoor air from the room / area / zone to facilitate air changes rate. This shall not to be confused with General Exhaust (GE) which primarily serves as Production Tools process exhaust in Fab building.

6.0 Performance Specification and System Provision

6.1 The General HVAC (GHVAC) equipment and components shall be as follows:

1. Air Handling Unit (AHU)
2. Makeup Air Unit (MAU)
3. GHVAC Outdoor Air or Exhaust Fans
4. Air Distribution Ductworks and grilles
5. Air Ductworks Control Dampers

*Note: Localized Fan Coil Units (FCU) may also be used for remote locations or where deemed necessary, but the preferred basis of design for HVAC configuration shall be Centralized Air Handling Units (AHU).

6.2 GHVAC System design for Electrical Rooms / Utilities Rooms shall generally be in line with the industrial recognized standards or guidelines (e.g. ASHRAE, LEED, etc) in terms of the criteria of energy performance and indoor air quality.

6.3 **System Design** – General Heating, Ventilating & Air Conditioning (GHVAC) systems are required to achieve the performance specified in this Standard. While the detail design is on a project-by-project basis, the following concept should be captured:

- 6.3.1 GHVAC System configuration shall consist of Central Air Handling Units (AHUs), Medium Pressure Supply Air Distribution Ductwork from AHU to Low Pressure Supply Air Distribution Ductwork within Electrical Rooms / Utilities Rooms, and Supply Grille Terminal Devices. Return Air shall be recirculated back to AHU through Return Air Grilles and Return Air Ductwork.
- 6.3.2 MAU Ventilation Airflow supply shall be designed accordingly to facilitate the required ventilation rate of the occupied areas. MAU Ventilation Airflow can be supplied directly to the central AHU units, where the Ventilation Airflow will mix with the Return Airflow. This mixed air will then be distributed as supply air to all the occupied areas.
- 6.3.3 AHU / MAU Systems shall be designed with ductworks & fittings (e.g. motorized dampers, sensing elements, control valves, etc) to have proper provision to ventilate, pressurize and regulate individual room air condition.
- 6.3.4 Fan motors should be designed with VSD to regulate the fan speed and airflow to operate at designated output

- corresponding to the required ventilation rate of each room (or zone).
- 6.3.5 AHU / MAU Systems design concept shall also be aligned with the smoke extraction strategy (i.e., de-smoke system) during the fire events or emergency cases, as required by local code or local jurisdictions.
- 6.3.6 Supply Air Distribution Ductwork shall be designed and completed with Motorized Dampers to ensure the air flowrate (and/or air pressure) can properly be balanced to achieve the design flowrate, based on the rooms or areas cooling load.
- 6.3.7 Mechanical Ventilation systems shall consist of Outdoor Air / Exhaust fans, Medium Pressure Air Distribution Ductwork, Low Pressure Air Distribution Ductwork within Electrical Rooms / Utilities Rooms, and Outdoor Air or Exhaust Air Grilles.
- 6.3.8 Mechanical Ventilation systems shall be designed with ductworks & fittings (e.g. motorized dampers, sensing elements, etc) to have proper provision to ventilate, pressurize and regulate individual room air condition. Mechanical Ventilation systems shall be aligned with the smoke extraction strategy (i.e. de-smoke system) during the fire events or emergency cases.
- 6.3.9 Since the utilities rooms are not typically occupied with permanent occupants, there may not be any required minimum amount of Outside Air Ventilation normally supplied to the space. Recommendation is to provide a minimum basic amount of Outdoor Air Ventilation to each space, or as required by Local Code and AHJ.
- 6.3.10 The Electrical Room / Utilities Room GHVAC Systems shall be designed and installed to achieve the required room condition and etc as outlined below; See **Table 1**.

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Table 1: Design Criteria of Electrical / Utilities Rooms Condition

Rooms / Areas	Ventilation mode (MV/AC)	Room Condition (Temperature, Relative Humidity)		Reference design cooling load, W/m ² <i>*Note 2</i>	Remarks
		°C	RH%		
HV/MV switchgear room w/o Transformer	AC	24 ± 2	≤70%	250 - 450	Recommended RH ≤65% <i>*Note 4</i>
LV Electrical & Control rooms	AC	24 ± 2	≤70%	300 - 450	Recommended RH ≤65% <i>*Note 4</i>
Electrical room c/w Transformer	AC	24 ± 2	≤70%	900 - 1,200	Recommended RH ≤65% <i>*Note 4</i>
Chemical control room / TGCM room	AC	24 ± 2	≤70%	200 - 400	Recommended RH ≤65% <i>*Note 4</i>
Server rooms, computer room, data center	AC	22 ± 2	30 - 65%	-	Cooling load to be determined with engineering calculation. To have N+1 and Generator emergency power back-up
MAU room	AC	≤28	-	120 - 200	
CDA room	AC	≤28	-	300 - 450	
Chiller room	AC	≤28	-	300 - 450	
UPW room	AC	≤28	-	250 - 350	
Scrubber room	AC / MV	≤28	-	-	
WWTP room	MV	-	-	-	
Reclaimed water room	MV	-	-	-	
Warehouse (FG: components, modules, SSDs)	AC	23 ± 5	50% ± 30%	-	Refer to the <u>Finished Goods Global RH, Temperature, and MST</u> for further detail.
Storage (Semi-FG: Die/wafers)	AC	23 ± 5	≤60%	-	Refer to the <u>Global Backend – Wafer Shelf Life for Micron Products</u> document for further detail.

Note 1: The GHVAC System shall be designed and installed with capability to meet the required room condition stated above, or in compliance with those required as recommended by Local AHJ, whichever is more stringent. The local climate profile shall also be taken into consideration to design an energy efficient system as recommended (or mandatory) by AHJ.

Note 2: The design cooling load shall include room cooling load and equipment cooling load during the normal operation. Engineering heat load calculation shall be performed to determine the actual design cooling load assigned to each room / area.

Note 3: The Mechanical Ventilation System shall be designed and installed to maintain the proper room condition for the operation of electrical equipment and installation inside the Electrical Room / Utility Room. The Mechanical Ventilation System shall be capable of maintaining the room air temperature and relative humidity within acceptable range to prevent the equipment overheat or condensation on the equipment surfaces.

Note 4: During the extended downtime due to preventive maintenance or corrective maintenance of GHVAC equipment, the room condition might be out of range during a short period of time (i.e., a few hours). The GHVAC equipment should be sized accordingly to be able to maintain the room temperature below a maximum of $\leq 28-35^{\circ}\text{C}$.

6.4 Performance Specification of GHVAC AHU / FCU / MAU

6.4.1 While the design of GHVAC AHU / FCU / MAU is highly dependent on the local weather profile, room load and equipment load requirements, the design parameters listed in **Table 2** shall serve as the guideline value to maintain the design consistency across Micron Electrical Rooms / Utilities Rooms.

Table 2: Performance Specification for GHVAC AHU / MAU

Parameter	Specification Requirement	Remarks
Supply Air Temperature	16 ~ 23°C (60.8 ~ 73.5 °F)	- Measured at outlet of AHU/FCU/MAU. - Subject to local atmospheric condition

Note: The AHU/MAU System shall be designed and installed with capability to meet the required performance specification stated above. Actual operational parameters should be considered on a case-by-case basis as per the local atmospheric condition, room load and equipment load requirements, as well as the actual ducting design.

7.0 Equipment Sizing & Design Requirements:

7.1 Redundancy – GHVAC System main equipment, namely AHU / MAU / FCU and Fans shall be designed:

- 7.1.1 For Electrical rooms, the number of AHUs serving a room or group of rooms (or zones) shall be minimum of (2) units at 75% capacity each. Therefore, when both units are operating simultaneously, they can operate at part load to provide 100% of the cooling capacity required. However, if 1 unit is offline for maintenance or failure downtime, there will still be 75% cooling capacity available, to temporarily maintain the room temperatures at the elevated temperatures of < 28-35°C. (Refer to Note 4 of section 6.3.10).
- 7.1.2 For Server rooms, Computer Rooms, and Data Centers, the number of AHUs serving a room or group of rooms (or zones) shall have one unit as N+1 duty standby.
- 7.1.3 For the other Utility Rooms, the number of AHUs serving a room or a group of rooms (or zones) shall be in the arrangement of N per the design capacity (with N being single unit or multiple units).
- 7.1.4 For exhaust systems that are required in order to meet requirements per the local code or local jurisdiction, there shall be one unit as N+1 duty standby.
- 7.1.5 The design redundancy shall be in compliance with the requirement of AHJ if it is more stringent than 7.1.1 to 7.1.4.

7.2 System Diversity and Safety Factor – GHVAC System sizing shall be selected so that the N capacity meets 100% of the system demand, space load and equipment load requirements at full build (after initial construction of the system), but also applying a safety factor of 15-20%.

- 7.2.1 To determine the system demand at N full build, the peak demand load shall be based on the Building's HVAC load calculation and the required equipment cooling. The estimation of HVAC load shall be considering, fenestration, infiltration, internal heat generation by occupants and equipment, the local weather profile, and etc.
- 7.2.2 Make-up air shall be considered accordingly to provide ventilation for occupants for indoor air quality requirements and in accordance with ASHRAE, compensate for the exhausted air, to maintain required room condition, as well as to regulate the room air pressure.
- 7.2.3 See **Table 3** for AHU Cooling / Heating Load criteria and calculation, **Table 4** for GHVAC Outdoor Air / Exhaust

system sizing criteria and calculation, and **Table 5** for AHU system sizing criteria and calculation.

Table 3: System Sizing Calculation – AHU System

AHU is the main HVAC component to treat the indoor air to maintain the Room Condition at the specified thermal comfort zone.

- Cooling Load Demand at Full Build, D_C , shall consider the following components of heat gain:

D_C = Heat gain through building envelopes + heat gain through fenestration + heat gain through outdoor air infiltration + heat load generated (sensible + latent) by operating equipment

- Heating Load Demand at Full Build, D_H , shall consider the following components of heat loss:

D_H = Heat Loss through building envelopes + heat loss through indoor air exfiltration

- AHU System N_C cooling capacity:
 $N = D_C \times 1.2$ (Safety Factor)
- AHU System N_H heating capacity:
 $N = D_H \times 1.2$ (Safety Factor)

Table 4: System Sizing Calculation – GHVAC Outdoor air / Exhaust System

With reference to Table 1

- GHVAC Outdoor air / Exhaust System Demand at Full Build, D :
 D = Total Outdoor air / Exhaust Flow Rate
- GHVAC Outdoor air / Exhaust System N capacity:
 $N = D \times 1.2$ (Safety Factor)

Table 5: System Sizing Calculation – MAU System

With reference to Table 1

- MAU System Demand at Full Build, D:
 $D = \text{Total Required Outdoor Air Flow Rate}$

**Actual MUA supply flowrate should take into consideration of the design air pressure level (i.e., relatively positive or negative) of each room / compartment. Air pressure level is typically maintained by having a different flowrate between make-up air and exhaust air.

- MAU System N capacity:
 $N = D_c \times 1.2 \text{ (Safety Factor)}$

**Actual MUA cooling / heating load calculation shall take into consideration of the outdoor air condition of the local weather profile.

7.3 **Equipment Arrangement** – It is recommended to use common, centralized GHVAC systems serving a group of rooms / areas as it is more cost and space effective, and reduces the total unit quantities required.

7.4 **Equipment Layout** – Equipment shall be suitably located to comply with all equipment maintenance clearances and access requirements. Adequate space and a clear routing path shall be reserved for future move-in of new equipment, and to allow move-in and move-out of large components during maintenance and repair.

7.4.1 Outdoor air / Exhaust Fans shall have adequate clearance space to allow for maintenance, repairs, and/or replacement on fan assembly, impeller, and motor.

7.4.2 AHU / MAU shall have adequate clearance space to allow for maintenance, repairs, and/or replacement on filters, coils, fan assembly, impeller, and motor.

7.4.3 Local FCU shall be located in accessible spaces above ceiling or on floor and shall have adequate clearance space to allow for maintenance, repairs, and/or replacement on filters, coils, fan assembly, impeller, and motor.

7.4.4 Equipment should be connected to a common header / plenum in such a way that minimizes the pressure drop, and to optimize the redundancy of the entire system.

7.5 Equipment Location - The location of the AHU / MAU / Ventilation Fans should be designed as such to reduce the length and pressure drop of the ductworks.

7.5.1 HVAC equipment shall be located away from Office areas / personnel workstations or areas for occupants' circulation to minimize interrupting the daily operation in these areas during the maintenance of the equipment.

7.5.2 HVAC equipment shall be located and installed as such to minimize the operating noise propagation into Office areas / personnel workstations.

7.5.3 The HVAC equipment should be accessible for the routine inspection and maintenance. The HVAC equipment is recommended to be in an indoor equipment room. The ductworks should be planned to avoid passing other rooms (or zones) to minimize the risk of spreading of hazards (e.g. fire / smoke).

7.6 Pressure Drop Calculations - Submit pressure drop calculations for each ductwork system, to show that each fan selection and the fan parameters (flow, pressure, power, etc.) are adequate to meet the design requirements.

7.7 Equipment Selection Criteria - AHU / FCU / MAU system selection criteria and requirements are as follow:

7.7.1 AHU / FCU shall consist of some of the following coils listed below, dependent on design location, climate, and load profile. The associated Coil water source shall be as follows:

1. Primary cooling coil: MTCHW

7.7.2 MAU shall consist of some of the following coils listed below, dependent on design location, climate, and load profile. The associated Coil water source shall be as follows:

1. Pre-heating coil (where required): Warm water
2. Primary cooling coil: MTCHW
3. Secondary cooling coil (where required): LTCHW
4. Re-heating coil (where required): Warm water

7.7.3 All coil sections shall have stainless steel sloped drain pans. From the cooling coil section on the interior shall be out of stainless steel. Face velocity through the unit approx. 2.5 m/s, through heat exchanger ≤ 2.5 m/s.

- 7.7.4 The units will have one access side for maintenance, repair and exchange of components. All components will be accessible via access doors with windows. The doors will have airtight gasket seals and have latches or hinges.
- 7.7.5 Each AHU or MAU shall have minimum 2 filter sections:
- Pre-filter: rated at 30 to 35% (MERV 5 or G3)
Recommend pressure drop: 65/130 Pa (clean/dirty)
 - Secondary filter: rated at 80 to 85% (MERV 13 or F7)
Recommend pressure drop: 150/250 Pa (clean/dirty)
- 7.7.6 Fan motor sizing and selection shall be as follows:
- Fans shall be Direct Drive where possible, instead of Belt Drive.
 - Fan motors shall be selected for non-overloading operation on all points of the performance curve at the design conditions.
 - Fan motors shall be rated for inverter duty with variable speed operation.
 - Fan motor efficiency rating shall be minimum IE3.
 - Fan motor IP rating shall be minimum IP55. If located outdoors, fan motor IP rating shall be IP56.
 - Fan shall have flexible ductwork connection on inlet side and flexible ductwork connection on discharge side.

7.8 Warehouse Temperature and Humidity Monitoring

- 7.8.1 Ceiling mounted Temperature and Humidity Sensors shall be used in warehouse for monitoring purpose.
- 7.8.2 Warehouse shall have at least 2 temperature and humidity sensors. (minimum 1 duty and 1 redundant T&H sensors) based on warehouse size.
- If the warehouse size is bigger than 1,000 m², the warehouse shall be separated into multiple temperature monitoring zones. Each zone size shall be approximately 500 m² each based on warehouse layout.
 - Each temperature monitoring zone shall have at least 1 temperature and humidity sensors.
- 7.8.3 Sensors shall be located away from ASRS system.
- 7.8.4 Sensors shall be located away from other obstruction and heat sources, such as ASRS panel, ductworks, utility piping, etc.
- 7.8.5 Sensors shall be located away from door facing external to prevent inaccurate reading.
- 7.8.6 Warehouse shall have pressurization of 5 Pa with respect to external area to minimize untreated air infiltration to warehouse.

8.0 Duct Design and Sizing Requirements:

- 8.1 **Ductwork Diversity and Sizing Factor** – The ductwork shall be sized according to the same sizing factors stated in the above Section 7.2 and below.
- 8.2 **Ductwork Mains Sizing Calculation** – Duct Mains shall be sized to accommodate of the airflow demand at full build (after initial construction of the system). See **Table 6**.

Table 6: Duct Mains Sizing Calculation

Ductwork to be sized to have its static pressure drop across all branches to be almost equal

- GHVAC Airflow Demand at Full Build, D:
D = Total Airflow Rate **

** Total airflow rate includes normal operating airflow and/or emergency ventilation airflow (e.g. make-up airflow for smoke extraction system, if applicable)

- GHVAC Duct Mains sizing:
Duct sizing = N+1+F; to include Future units

- 8.3 **Duct Sizing Criteria** – Main Duct Sizing shall be selected based on the full build capacity, including the future equipment (N+1+F). It is recommended for the duct sizing to be designed with maximum specific pressure drop not to exceed the following:

8.3.1 Low Pressure Supply Air Duct – 0.6 Pa/m

8.3.2 Medium Pressure Supply Air Duct – 0.8~1.0 Pa/m

- 8.4 **Duct Pressure Rating**, – Ductwork mains shall be designed to meet the minimum pressure class in accordance with SMACNA Standards, such as +3 inches w.g. (+750 Pa). If the system pressure at the main ductwork exceeds this value, then the ductwork pressure rating shall meet or exceed the system pressure.

8.5 Duct Supply / Return / Exhaust Air Grilles – The Air Grilles should evenly be distributed to have an effective air distribution & recirculation within the room.

8.5.1 Supply Air Grilles and Return Air Grilles should be positioned at the top part of the room.

8.5.2 Supply Air Grilles and Return Air Grilles should be strategically positioned to enhance mixing of the conditioned air with the room air. Position of these grilles should avoid air path short-circuit.

8.5.3 Return / Exhaust Air Grilles should generally be positioned at the top part of the room and to avoid prematurely exhausting the supply air from the Supply Air Grille.

8.6 Duct Outdoor Air intake / Exhaust Air Outlet – Outdoor air intakes shall not be within 5m from any exhaust discharge openings / outlets.

8.6.1 The distance from an air intake to a cooling tower is measured from the base of the cooling tower.

8.6.2 Outdoor air intakes shall not be within 10m from any exhaust discharge openings / outlets of noxious or potentially hazardous exhaust.

8.7 Materials of Construction - GHVAC Ductwork, rectangular and circular, shall be made of Galvanized Steel. Ductwork to be insulated from the equipment unit outlet connection up to the supply air grille in the rooms. The ductwork insulation material shall be vapor-tight, with a closed cell structure, made of expanded cellular polyethylene foam

- **Gaskets:** Neoprene tape, silicon-free gaskets shall be used for GHVAC ductwork.

8.8 Ductwork Installation - Low pressure rectangular ductwork shall be made of galvanized steel sheet, in accordance with SMACNA requirements. The ductwork system shall be complete with transitions, bends, tees, supports, dampers, offsets, flexible connections, take-offs and similar fittings necessary for the balancing and full operation of the air distribution system. Ductwork shall be installed in accordance with SMACNA "Industrial Duct Construction Standards."

- No flexible duct connection should be used for Electrical Rooms / Utilities Rooms

8.9 Fire Dampers and Fire-rated enclosure

- 8.9.1 Fire Dampers shall be installed in ductworks wherever there is penetration of fire compartment.
- 8.9.2 If Fire Dampers cannot be installed in ductworks due to its nature of operation (e.g. generator flue gas exhaust, etc), fire-rated enclosure on the ductworks should be considered as the alternative provision to maintain the fire compartment integrity.

Note: If the local code or jurisdiction prohibits the installation fire dampers in the ductwork, it shall take precedence.

9.0 **Other Accessories** - Provide other accessories as outlined in **Table 7**.

Table 7: Accessories to be installed

Accessories	Type & Installation Location
Insulation	- Ductworks shall be insulated wherever there is potential occurrence of air condensation (internal or external of ductworks) - Install for SA ductworks - Install for all CHW piping - Install for refrigerant piping - Install for condensate drain piping
Ductwork Supports	- Provide duct hanger and support material gauges in compliance with details on the Drawings. Where details are not provided, comply with SMACNA Industrial Duct Construction Standards. - Ductwork hanger and suspension material may be threaded rod, structural shapes, or strut material. The use of wire as suspension material is prohibited.
Piping Supports	- Install piping hangers, supports, racks, and clamps at the appropriate spacing interval as required by the system and piping load and layout - Provide seismic restraints and supports as deemed required during design, based on site location and local jurisdiction requirements - Provide vibration isolation for piping supports as deemed required during design, based on the building vibration criteria
Pressure Gauges	- Install at cooler supply and return connection for CHW - Install at Heater supply and return connection for WW
Temperature Gauges	- Install at cooler supply and return connection for CHW - Install at Heater supply and return connection for WW
Pressure Difference Gauges	Install at each filter section
Air Vents	- Install at cooler supply and return connection for CHW - Install at Heater supply and return connection for WW

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Accessories	Type & Installation Location
Drain Ports	• Install at all low point locations
Valves	• All butterfly valves shall be rated for "bi- directional" duty with positive shutoff in both directions • All butterfly valves shall be lug-style
Control Valves	• Provide control valve type and quantity as outlined in this standard
Labeling and Identification	• Provide identification labels at minimum of every 6m distance and change of direction • Refer to the specific project's Construction Specification, (i.e. Spec Section 15190 or equivalent Spec) for additional requirements. <u>Ductwork</u> • "SA" nomenclature shall be used for air-conditioning Supply Air • "RA" nomenclature shall be used for air-conditioning Return Air • "MAU" nomenclature shall be used for air-conditioning Make-Up air • "GHVAC Exhaust" nomenclature shall be used for General HVAC Exhaust

10.0 Sustainability Design:

- 10.1 **Ductwork Pressure Drop Sizing** – It is recommended to increase the ductwork sizes where possible, to minimize the air velocity and pressure drop through the ductwork. This will result in lower fan energy consumption.
- 10.2 **LEED Design** – GHVAC System design in line with LEED guideline recommendation will enhance the overall system sustainability in terms of energy efficiency and indoor environmental quality (IEQ).

11.0 Electrical Design:

- 11.1 Electrical power supply system for all GHVAC equipment shall be designed and installed in compliance with **Global Facilities - Design & Construction Standard - Electrical - Electrical Design Specification**.
- 11.2 **Power Segregation** – Electrical power supply for GHVAC equipment shall be segregated among a minimum of 2 separate upstream power sources (wherever possible). The quantity of equipment served from each upstream power source shall be split evenly
- 11.3 **GHVAC system and system controls** used for Electrical Rooms / Utilities Rooms cooling and ventilation shall not require Emergency Power or Un-interrupted Power (UPS).
- 11.4 **GHVAC system and system controls** used for Computer Room / Server Room / Data Center cooling and ventilation shall be provided with Generator Emergency Power Back-up and Un-interrupted Power (UPS).
- 11.5 **GHVAC system and system controls** used for Life Safety Systems and Emergency Systems shall be connected to Emergency Power and/or Un-interrupted Power (UPS), as required by Global Standards or Local Code Regulations.
- 11.6 **Variable Frequency Drives** – All central AHU and MAU fans shall be VFD driven for variable speed operation. Variable Frequency Drives shall be selected to be one size larger than the power rating of the actual fan motor size. Local FCU fans may not require VFDs.
 - 11.6.1 Variable Frequency Drives shall be compliant with SEMI standard F47 for voltage sag immunity, to provide ride-through capability during momentary power interruption or voltage sag.
 - 11.6.2 Variable Frequency Drives shall be configured to “Fail Last Speed” configuration upon any failure, including but not limited to, loss of power, loss of communications, etc.
- 11.7 **Motors** - Fan motors shall be inverter duty rated with insulated bearings and shaft grounding kits. Motors shall be compatible with the associated VFD.
 - 11.7.1 All motors located outdoors shall be IP56 rated protection.
 - 11.7.2 All motors located indoors shall be IP55 rated protection.
 - 11.7.3 All motors shall be National Electrical Manufacturers Association (NEMA) IE3 rated or better.

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- 11.8 Local Disconnect Switches** – Each fan as part of the AHU / MAU / FCU unit shall have a lockable type, local disconnect switch installed adjacent to the fan, to allow for manual isolation and lockout/tagout (LOTO/COHE) of the fan. All disconnect switches must be reasonably accessible for the authorized personnel to safely isolate the power supply to facilitate maintenance works.

12.0 Instrumentation:

- 12.1 The following Instrumentation shall be installed and monitored/controlled Avia FMCS: See **Table 9**.

Table 8 - Instrumentation for AHU / MAU

Instrumentation	Type and Installation Location	
	AHU / FCU	MAU
Pressure Transmitters	Install at AHU Common Main Supply header (Quantity: 1 duty)	Install at Main header (Quantity: 1 duty)
Differential Pressure gauges	Install at each filter bank (Quantity: 1 duty)	Install at each filter bank (Quantity: 1 duty)
Temperature / Humidity Transmitters	Install at inlet and outlet of AHU / FCU (Quantity: 2 duty Temperature transmitter)	- Install at outlet of MAU (Quantity: 1 duty Temperature transmitter, 1 duty Humidity transmitter) - Can be converted to Dewpoint & Enthalpy
Temperature Transmitter (in unit)	Nil	Install at outlet of heating and Cooling coils (Quantity: 1 duty each)
Room HVAC (Temperature / Humidity Transmitter)	Room shall have air temperature and humidity sensors for SCADA monitoring and control purposes.	

- 12.2 Instrumentation points of connection shall be provided with a means of isolation (ball valve, etc.) for maintenance and/or replacement.
- 12.3 Smoke detectors / Instruments that are monitoring for smoke shall be located as such that the detection signal capable of identifying the specific zone of the incident (e.g., in the Return Air Duct), and in

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accordance with local code or local jurisdiction requirements. AHUs serving the specific zone of the incident shall be stopped if the smoke detector is triggered to prevent the smoke spreading into other areas.

- 12.4 The Pressure Differential Transmitters on ductwork shall reference the Zero Reference Pipe (ZRP). This shall be the same ZRP used for Make-up air unit pressurization reference.

12.4.1 ZRP shall reference outdoor atmospheric pressure. ZRP external probe shall be located in a place with minimum wind interference and shall have adequate cover protection from wind.

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13.0 Control Sequence Philosophy:

- 13.1 The overall philosophy for system control and operation shall be listed in this section. See **Table 9**.

Table 9: System Control Logic

Control Parameter	Control Logic
GHVAC Ventilation Fans Start/Stop	- Ventilation Fans Start/Stop shall be done automatically via SCADA or hardwired logic, based on required number of running fans - Ventilation Fans Start/Stop/Rev shall be capable of being remotely controlled via SCADA by Operation Personnel.
AHU / FCU	- AHU & FCU Start/Stop shall be done automatically via SCADA based on required number of running units VSD Speed of AHU Fans shall be modulated to maintain the pressure setpoint at the Main duct. VSD speed of FCU fans (if applicable) shall be modulated based on the room load and equipment load requirements. Cooling - Position output of 1st cooling coil temperature control valve shall be modulated to maintain the temperature setpoint at the control point (which can be the Return Air Duct, Supply Air Main Header, Room Temperature sensor, etc).
MAU	- MAU Start/Stop shall be done automatically via SCADA based on required number of running units VSD Speed of MAU Fans shall be modulated to maintain the pressure setpoint at the Main duct. Preheating - Preheating coil (if necessary) temperature control valve shall be modulated to pre-heat the outdoor air to the temperature set point at outlet of preheating coil. Humidifier - Humidifier (if necessary) control valve shall be modulated to the humidity set point at the outlet of the humidifier. Cooling / Dehumidification

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Control Parameter	Control Logic
	- Position output of 1st cooling coil temperature control valve shall be modulated to maintain the temperature (and dewpoint) setpoint at outlet of 1st cooling coil. Reheating - Position output of reheat coil (if necessary) control valves shall be modulated to maintain the relative humidity (RH%) setpoint at outlet of MAU.
Ductwork Motorized Dampers	- Motorized dampers opening percentage to be measured and confirmed through an air-balancing activity. - Motorized dampers opening percentage to be actuated to rated percentage, either during Normal or Emergency mode, to allowed for proper air flowrate. - Normal / Emergency mode rated flowrate to be validated by manual measurement.

13.2 A detailed Sequence of Operation (SOO) shall be developed specifically for each system, as part of each construction project.

13.3 Control Valves and Dampers

13.3.1 The following control valve and damper fail positions shall be configured. See **Table 10**

Table 10: GHVAC System Motorized Dampers and Control Valves

Control Valve or Damper	Control Valve or Damper Fail Position		
	Loss of Air	Loss of Electricity	Loss of Control Signal
AHU / FCU / MAU Ductworks Motorized Dampers	Fail Last State	Fail Last State	Fail Last State
AHU / FCU / MAU Temperature Control Valves	Fail Last State	Fail Last State	Fail Last State

13.3.2 The control valves and dampers shall be double-acting type actuators, capable of failing to last state position and holding the position for unlimited duration without needing control air, power, or PLC signal.

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- 13.3.3 After a loss event of control air, power, or PLC signal, and after it is restored, the control valve or damper output command/position shall be set to the same command/position it was operating at prior to the loss event.
- 13.3.4 All control valves and dampers shall have a manual override bypass, to allow manual operation of the valve or damper upon a failure.
- 13.3.5 For motorized dampers with actuators, it is recommended to have square-shaped shafts connecting the damper and the actuator, to prevent slippage and mis-operation. Prior to system startup, perform inspection and operations check of all dampers and actuators including shaft, linkages, and connection points.

14.0 Inter-Discipline Coordination Matrix:

14.1 The following provisions shall be provided by each of the following disciplines and coordinated accordingly: See **Table 11**.

Table 11: Inter-Discipline Coordination Matrix

Discipline	Provision
Architectural	• Provide Mechanical Room space for GHVAC Equipment. • All rooms shall be constructed with appropriate fire compartmentation and fire rated partition / wall. • Cut openings in walls and floors for ducting penetrations, and seal openings after duct installation • Provide adequate fire stopping at all required fire wall penetrations • Provide sloped floors and floor drains in Mechanical Room space containing the GHVAC equipment.
Mechanical	• Provide GHVAC space conditioning for Mechanical Room space containing the GHVAC equipment. Room temperature shall be designed to maintain the space temperature range of 24°C to 28°C. • Provide heating water, chilled water and city water supply piping, and accessories as outlined in Standards, to provide sufficient heating, cooling water for GHVAC equipment • Provide drain water piping, floor drains and accessories to be located suitably to allow proper draining of equipment, condensate drains and auto- drains, piping, and air vents • Louver shall be constructed to blend in Architectural façade design, e.g. color, structure, etc. • Exhaust louver shall be constructed at least 5 meter away from any Outdoor air intake / building opening. • Outdoor air louver shall be constructed at least 5 meter away from any exhaust louver. • Fire dampers shall be installed in ductworks wherever it penetrates the fire compartment.
Electrical	• Provide local disconnect switches for all fan motors. • Provide power wiring for all Electrical equipment/ components • Provide grounding for all Electrical equipment/ components • Provide Emergency Power / UPS Power as outlined in this Standard • Provide Power Segregation as outlined in this Standard • Provide VFDs for all fan motors. • Provide lightning protection for outdoor equipment including exhaust discharge stacks.

Discipline	Provision
Controls	• Provide control damper and control valve actuators and positioners • Provide control wiring • Provide instrument control air tubing • Provide Zero Reference Pipe (ZRP) for atmospheric pressure reference of Exhaust and Makeup Air Systems • Provide FMCS/SCADA system monitoring and controls of all instrumentation and transmitters including pressure, temperature, flow, gas detector, fire detector etc. • Provide FMCS/SCADA system connectivity of all Local Control Panels

15.0 Quality Inspection and Testing:

15.1 **Inspection and Test Plan (ITP)** – Create formal Inspection and Test Plans (ITP) that outline all requirements for quality verification, inspections, testing, and commissioning:

15.1.1 Piping and Ductwork

- Piping and Ductwork Shop Drawings and Submittals
- Incoming Material Inspection
- Material Storage
- Assembly Procedures, Qualifications, and Inspection
- Duct Supports Installation and Verification
- QA/QC Inspection & Punchlist
- Ductwork Leak Testing –At least 20% HVAC ductwork shall be leak tested.
- Leak Testing and Pressure Testing
- Condensate piping installation shall be maintained with proper gradient to ensure the gravity flow is sufficient to drain out the condensate.
- Quality Certification
- Final QA/QC Inspection & Punchlist
- Training, Documentation, and Handover

15.1.2 Equipment

- Equipment Shop Drawings and Submittals
- Factory Acceptance Test
- Incoming Equipment Inspection
- Equipment Move-In and Installation
- QA/QC Inspection & Punchlist
- Equipment Pre-Startup Checks – Mechanical, Electrical, Controls, etc.

- Operational Acceptance Test (OAT) - Standalone Equipment Startup per Vendor's Procedure to allow equipment to run in MANUAL mode. Also considered "Early Run" Equipment.

15.1.3 System

- Site Acceptance Test (SAT) – Test System operation and control functions in accordance with P&ID, SOO, and Specifications.
- Integrated System Test (IST) – Test full System operation including all system interdependencies, interlocks, and critical failure modes.
- Final QA/QC Inspection & Punchlist
- Training, Documentation, and Handover

16.0 Startup Commissioning / Site Acceptance Test:

- 16.1 At a minimum, the following system control functions shall be tested and successfully proven during the System Startup Testing & Commissioning / Acceptance Tests.

16.1.1 Fans (including for AHU/MAU/FCU/VAV/Exhaust & Ventilation Fans)

- Fan Motor Rotation Check
- Fan/Motor Alignment Check and Lubrication
- All Damper, Valve, and Actuator Checks
- Motorized Damper operation checks including actuator, shaft, linkages, and connection points
- VSD Settings and Functionality Checks
- Control interface and communication checks to FMCS/SCADA
- Fan Standalone Startup Procedure per Manufacturer's procedure
- Fan Vibration Level Checks
- Verify Fans Operation through its operating range (0 to 50/60 Hz)
- Verify Fan Start/Stop Function to Rotate Operating Fan
- Verify Fan Auto-Start sequencing, upon failure of an operating Fan
- Verify VSD fails last speed upon any failure
- Verify control interface and communication checks to FMCS/SCADA
- Control interface and communication checks to Local Control Panel, PLC, and FMCS/SCADA.

16.1.2 AHU / FCU / MAU / FCU / VAV

- Confirm the external static pressure and the flowrate
- Confirm system pressure control is stable during fan manual start/stop rotation, and fan auto-start procedure
- Confirm Auto-switch to backup pressure transmitter(s) upon failure of main pressure transmitter(s)
- Confirm all control valves and dampers fail to the stated position per this Standard, upon loss of instrument air, power, and/or control signal.
- Verify all dampers and coil control valves operation through its operating range (0-100%)
- Confirm all unit temperature sensors and freeze stat operation
- Confirm all UV light and Bi-Polar Ionization function and operation
- Confirm all smoke detectors operation and interlock functions
- Confirm space temperature, humidity, and CO2 sensors monitoring and operation
- Air flow test and balancing for all rooms, VAV boxes, duct branches, SA diffusers & RA grilles and exhaust grilles.
- Verify all alarms are enabled and activated in SCADA
- Verify equipment on emergency power is able to operate on emergency power, after simulated loss of normal power
- Perform Comprehensive Power loss and recovery (blackout) test

16.1.3 Integrated system performance test

- Confirm system pressure control is stable during fan manual start/stop rotation, and fan auto-start procedure
- Confirm Auto-switch to backup pressure transmitter(s) upon failure of main pressure transmitter(s)
- Confirm all control valves and dampers fail to the stated position per this Standard, upon loss of instrument air, power, and/or control signal.
- Verify all alarms are enabled and activated in SCADA
- Verify equipment on emergency power is able to operate on emergency power, after simulated loss of normal power
- Perform Comprehensive Power loss and recovery (blackout) test
- system integrated testing, AHU / MAU / Exhaust System and control / interlock with motorized dampers activation.

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17.0 Reference Specifications:

- 17.1 The following Micron Global Construction Specifications shall be referenced during design, engineering, and construction. The Specifications shown below are in accordance with the Construction Specifications Institute (CSI) Master Format; Refer to the specific Construction Specifications for each project. See **Table 13**.

Table 13: Specification Sections

Spec Section	Spec Contents
Section 230519	Meters and Gauges for HVAC Piping
Section 230529	Hangers and Supports for Piping
Section 230548.13	Vibration Isolation for HVAC
Section 230553	Mechanical Identification
Section 230719	HVAC Piping Insulation
Section 230713	Ducting Insulation
Section 232113	Pipe and Pipe Fittings
Section 233413	Axial Fans
Section 233416.13	FRP Centrifugal Fans
Section 238219	Fan Coil Units
Section 432100	HVAC Centrifugal Pumps

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18.0 Document Control

18.1 Approval: GLOBAL_FAC_GFTT_APPR

18.2 Notification:

GLOBAL_FAC_MANAGERS
GLOBAL_FAC_COMMUNICATIONS
GLOBAL_FAC_CONSTRUCTION
GLOBAL_FAC_NOTIFY
FCG_F4_FAC_NOTIFY
FCG_F6_FAC_NOTIFY
FCG_F10_FAC_NOTIFY
FCG_F11_FAC_NOTIFY
FCG_F15_FAC_NOTIFY
FCG_F16_FAC_NOTIFY
FCG_MMY_FAC_NOTIFY
FCG_MMP_FAC_NOTIFY
FCG_MSB_FAC_NOTIFY
FCG_MTB_FAC_NOTIFY
FCG_MXA_FAC_NOTIFY

18.3 Final Viewer Access

All Micron Team Members at all Sites

18.4 Retention

Adherence to the requirements specified in the
[Global Facilities – Document Control Procedures](#)

18.5 Review

Biennially (two years)

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19.0 Revision History

Revision #	Section Changed	Details of Changes	Date
0	NA	New document	05/06/2022
1	All	Add warehouse design standard and update document title.	12/12/2023
	Section 3.0	Updated Section 3.0 Definitions and Acronyms to cater the re-organization	
	Section 4.0	Updated Section 4.0 Roles and Responsibilities to cater the re-organization.	
	Section 5.0	Added 5.4	
	Section 6.3.10, Table 1	Added Warehouse (FG: components, modules, SSDs) and Storage (Semi FG: die/wafers)	
	Section 7.0	Added 7.8	
	Section 17.1.	Updated with Micron Global Construction Specification.	
	Section 17.1, Table 13	Updated with Micron Global Construction Specification.	05/02/2024
2	3.0	Remove Acronym of FCT, GFC, GFTT, Global Facilities Central Team, Construction Team, Operations Support Team and planning Team.	
	Table 1	Note 2 Remove "The design cooling load shall not be lower than the Reference Design Cooling Load listed above to ensure design consistency across network sites"	